

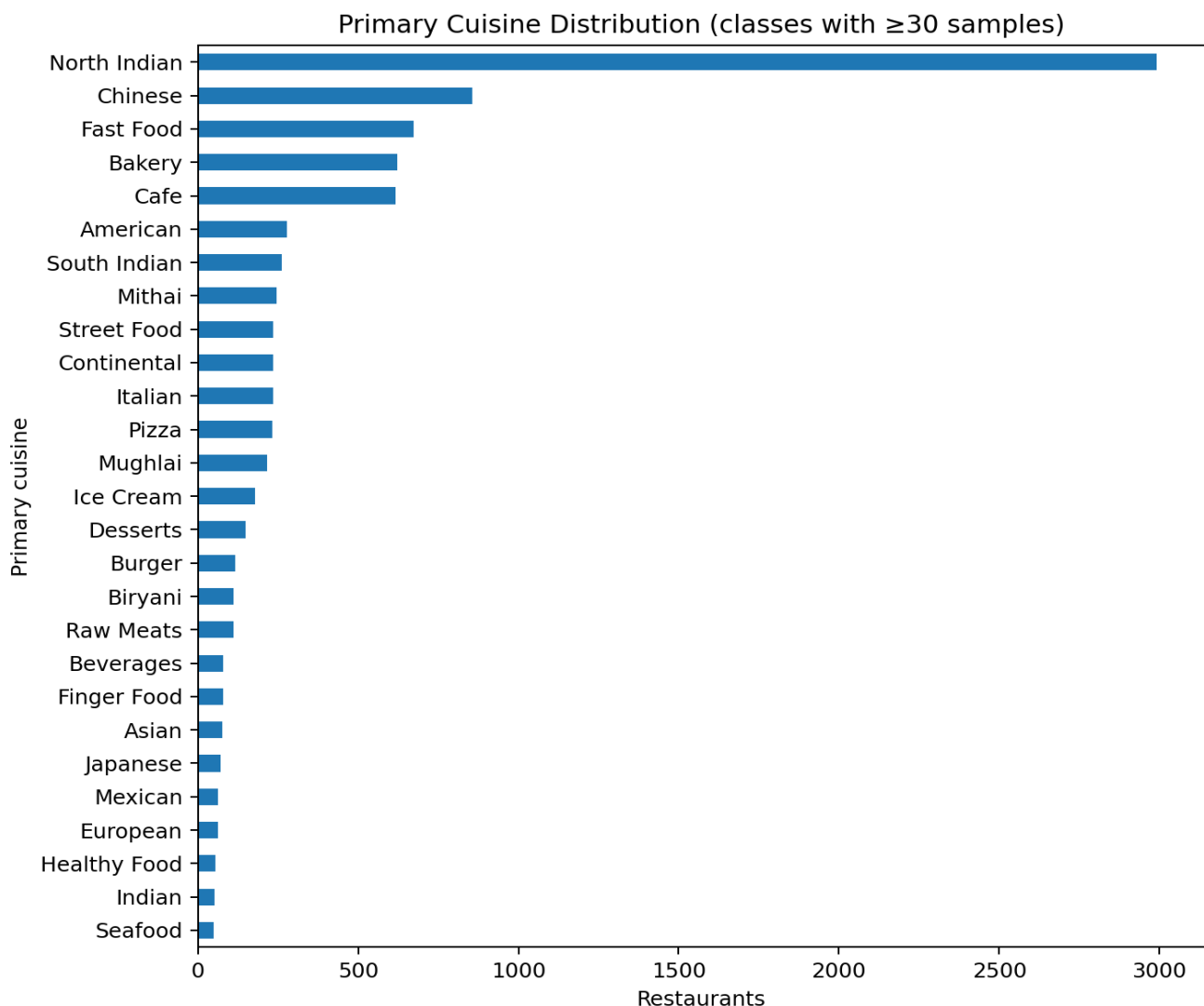
Cognifyz Technologies — Machine Learning Internship

Task 3: Cuisine Classification

Objective: Develop a machine learning model to classify restaurants based on their cuisines, following the internship task requirements.

1. Problem Formulation

The raw Cuisine field contains multi-cuisine strings. For a standard single-label multiclass benchmark, the first listed cuisine is treated as the **Primary Cuisine**. Cuisine classes with fewer than 30 examples were excluded because extremely rare labels cannot support a meaningful stratified train/test benchmark. This choice is explicitly documented.



2. Leakage Prevention & Preprocessing

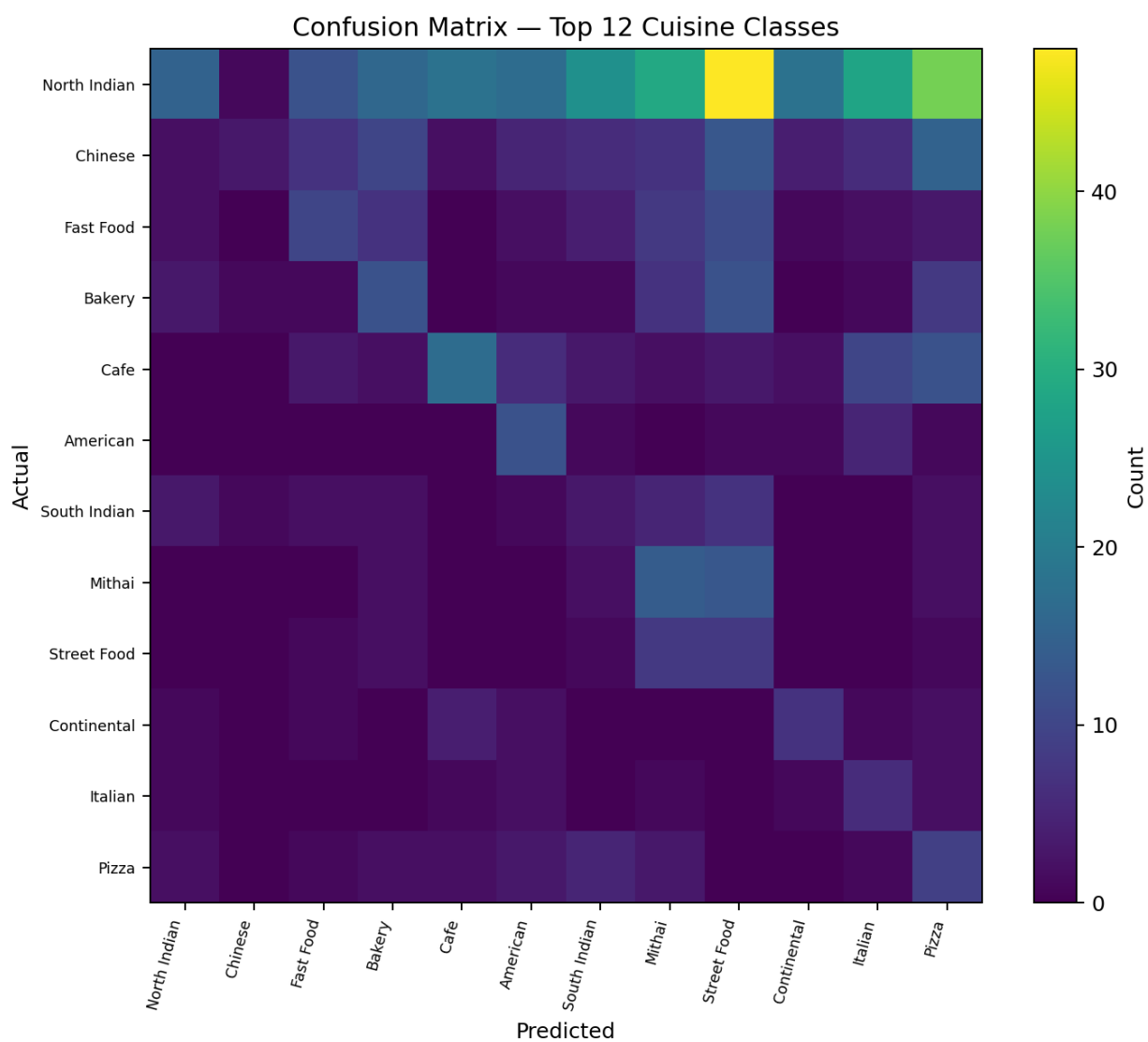
The original Cuisines field is not used as an input because it defines the target. City and Locality are represented with TF-IDF features; price range and service fields are tokenized as additional features. The split is stratified and reproducible.

3. Model

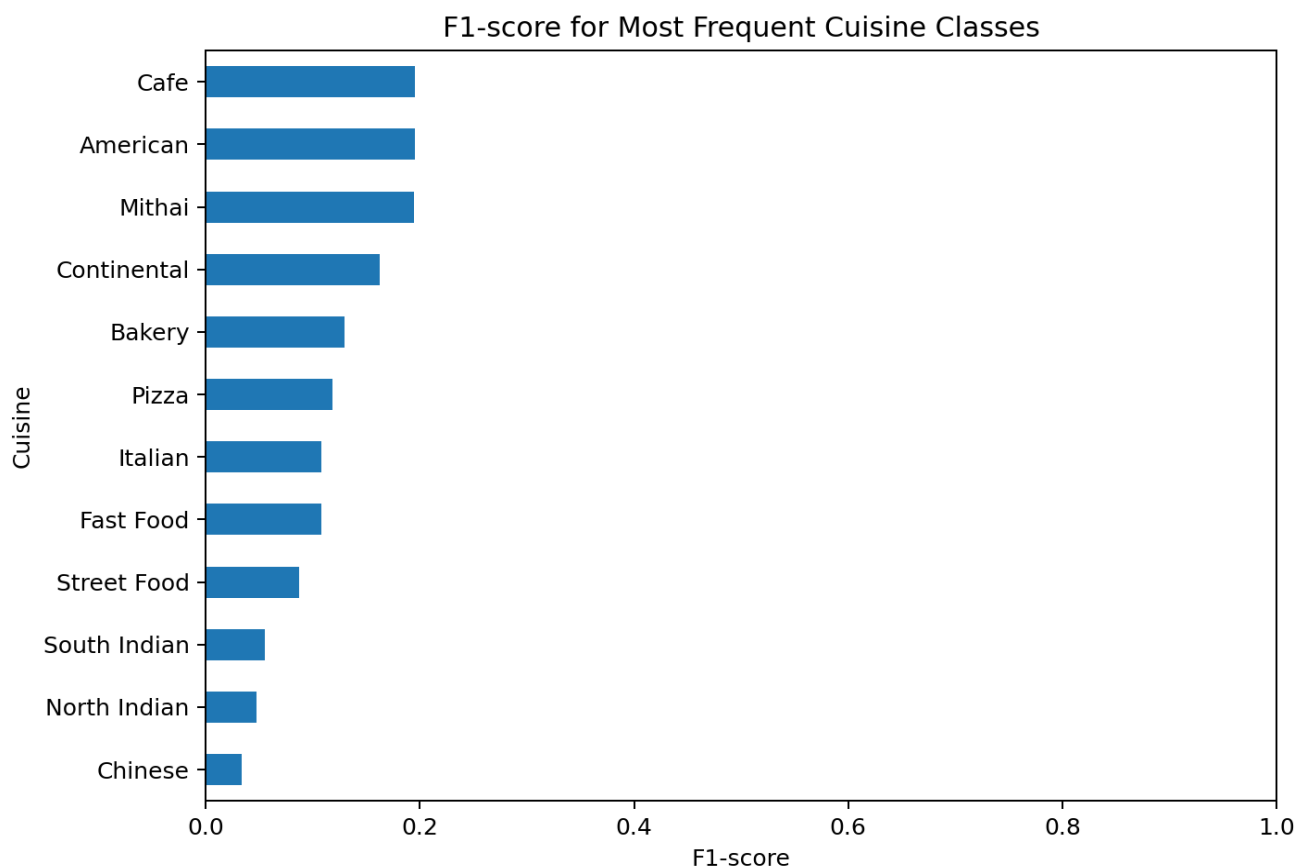
A Logistic Regression classifier with balanced class weights was selected for high-dimensional sparse TF-IDF features. Balanced weighting reduces the tendency to favor the most frequent cuisine classes.

4. Evaluation

Metric	Result
Accuracy	0.091 (9.1%)
Weighted Precision	0.304
Weighted Recall	0.091
Weighted F1	0.088
Macro F1	0.100
Majority baseline	0.335 (33.5%)
Top-3 accuracy	0.279 (27.9%)

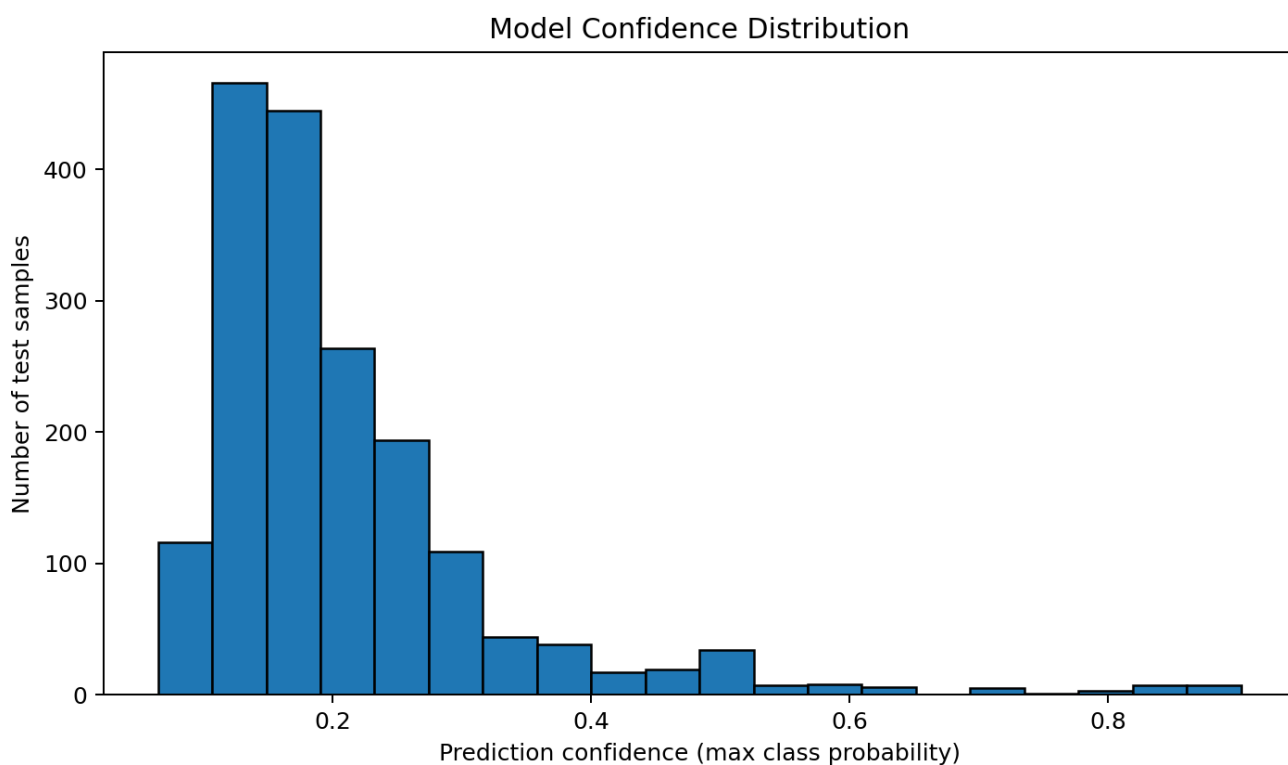


5. Per-Class Performance



Overall accuracy can hide weak performance on less common classes, so macro F1 and class-level F1 are reported alongside weighted metrics.

6. Model Confidence



Maximum predicted probability indicates model confidence, not guaranteed correctness; calibration would be needed for production decisions.

7. Engineering Limitations & Next Steps

- Cuisine is naturally multi-label in the source data; using the first listed cuisine creates a practical single-label benchmark.
- The dataset lacks rich menu/review text, limiting semantic cuisine classification.
- A production system could use true multi-label classification, multilingual embeddings or a hierarchical cuisine taxonomy.
- Cross-validation and deeper class-wise error analysis would strengthen model validation.
- Top-K cuisine predictions can be exposed instead of forcing a single answer.

Prepared for Cognifyz Technologies — Machine Learning Internship | Task 3